

School Of Computing and Informatics

Game Design and Development

Lost Signal

Spring – 2026

Table of Contents

1. Theoretical Foundation ..... **Error! Bookmark not defined.**

1.1 three different game engines currently popular in the industry .. **Error! Bookmark not defined.**

1.1.1 Unity ..... **Error! Bookmark not defined.**

1.1.2 Unreal Engine ..... **Error! Bookmark not defined.**

1.1.3 CryEngine ..... **Error! Bookmark not defined.**

1.1.3 Comparison ..... **Error! Bookmark not defined.**

1.2 Origins & Evolution of Game Engines ..... **Error! Bookmark not defined.**

1.2.1 1970s & early 1980s ..... **Error! Bookmark not defined.**

1.2.2 Late 1980s & 1990s ..... **Error! Bookmark not defined.**

1.2.3 2000s ..... **Error! Bookmark not defined.**

1.2.4 2010s ..... **Error! Bookmark not defined.**

1.3 Future Trends in Game Engine Development ..... **Error! Bookmark not defined.**

1.3.1 Cloud Computing ..... **Error! Bookmark not defined.**

1.3.2 Procedural Generation ..... **Error! Bookmark not defined.**

1.3.3 Virtual Reality ..... **Error! Bookmark not defined.**

1.3.4 Predict the Future ..... **Error! Bookmark not defined.**

1.4 Role of AI and ML in Modern Game Development ..... **Error! Bookmark not defined.**

1.4.1 Gameplay Experience ..... **Error! Bookmark not defined.**

1.4.2 Character Behaviours ..... **Error! Bookmark not defined.**

1.4.3 Procedural Content Generation ..... **Error! Bookmark not defined.**

1.5 Review of Published Unity Games ..... **Error! Bookmark not defined.**

1.5.1 Among US ..... **Error! Bookmark not defined.**

1.5.2 Rust ..... **Error! Bookmark not defined.**

2. Design Blueprint 3

2.1 Understanding Game Design Document 3

2.2 Game Design Document 3

2.2.1 Overview 4

Concept 4

Target Audience 4

2.2.2 Gameplay 4

Core Loop 4

Mechanics 4

2.2.3 Story & Levels 5

Story 5

Levels 5

2.2.4 Characters 6
2.2.5 Technical Specifications 6
2.2.6 Sound Design 6
2.2.7 Milestones 6
2.2.8 Prioritized Features 6
3. Technical Implementation 7
3.2 Unity Features 7
3.2.1 Prefabs 7
3.2.2 Physics engine 7
3.2.3 Animation System 8
3.2.4 UI System 8
3.2.5 Audio System 9
3.2.6 Scene Management 9
3.3 Challenges 9
Implementing & Debugging Mechanics 9
Assets 10
Game Feel 10
3.4 Asset Management and Integration 10
3.5 Implementation of GDD Features 11
Mechanics 11
Story & Levels 11
Characters 12
Technical Specification 12
Sound Design 12
Overall 12
4. Validation 12
4.1 Playtesting & Feedback Collection 12
4.2 Interpreting Peer Review Feedback 13
Testing Log 13
Suggestions Log 13
Strengths, weaknesses, and Opportunities 13
5. Refinement & Future Plan 14
References 14
Tutorials, Forums, Documentations 16
Speech Generation Platforms 17
Assets 17

1. Design Blueprint

1.1 Understanding Game Design Document

After reviewing several Game Design Documents and templates I understood that Game Design Document is vital for the game creation as it considered as the backbone of creating the game and it includes most of the details and features about the game idea, and how this game will be created which heavily support and guides the process of creating the game.

The main Game Design Document I reviewed was An Ant's Life GDD which is describe a simulation game that places the player inside the life of an ant. It covers:

- Game Overview and Introduction
- Game Structure including controls and scoring systems
- Ant-Human Interface including senses and actions like movement
- Universe simulation particles that represent chemicals
- Game Elements such as NPC'S
- Levels
- Sixty Seconds of Play describing the story and player experience.

Reflect briefly on what was useful and what you found limiting in the ready-made GDD

An Ant's Life GDD guided my thinking in different ways, and the most important part that I learned from it that GDD should explain the developed game, as each section of the GDD should clearly explain itself, for example "Sixty seconds of Play" section was very helpful because it showed how a player experience could be planned before development starts, and it made me start thinking differently, as at first, the game was minimal in my head with no good player experience or story, but then I expanded my thinking to think about how should the player experience be and how the story should be written. Also, the GDD provided by unity guided me in providing a good GDD structure that helped me organize my ideas into categories.

I borrowed the idea where I can structure each level around a different challenge that builds on the challenge before, as in An Ant's life, each level introduces a new phase of the game with new tasks and conditions, and this was slightly borrowed in the level structure of Lost Signal GDD. An Ant's life has some limitations such as it was designed for a simulation game, while Lost Signal is an educational survival game as some mechanics could not be directly adapted to Lost Signal. Also, it did not say any technical details like the platform, and game engine which is something I would like to know before installing a game.

1.2 Game Design Document

The process started of deciding which educational field should the game focus on, so I choose Physics (Gravity) as the main field that the game should focus on. Then I

reviewed existing GDD's and started to think how to make the game enjoyable and not only provide a traditional learning experience but also an enjoyable experience.

The first idea was simple and minimal, but then I focused on making the game enjoyable through making a story around the game and there should be a goal that the player should reach

1.2.1 Overview

Concept

Lost Signal is a 3D first-person educational, and survival game that places the player on an alien planet, which he faces a trial created by an alien boss. Captain John must survive those trials through radio communication with Emilie, as he must avoid falling objects, avoid and fight aliens in the trial environment, reach the spaceship, rescue Emilie, and finally escape the planet. The game taches gravity through falling objects, mass through different behaviours of falling objects, projectile motion through throwing objects on aliens, while Emilie is explaining these physics concept through radio communication.

Target Audience

- Players that are +12 including:
    - Teenagers
    - Students
    - Players who like sci-fi and educational games
    - Players who like survival games

1.2.2 Gameplay

Core Loop

The core loop of Lost Signal is Listen → Explore → Survive → Progress.

The player will listen to guidance from captain Emilie through radio communication where she guides him and teach him at the same time. Then the player will explore the trial environment while surviving falling objects, and surviving aliens by avoiding them or spawn and throw rocks on them. Then the player reahc the end of the trial and progress to the next one where.

Mechanics

There will be different mechanics and systems in the game such as:

- Movement: The player can walk or run and look around while experiencing the trials.
- Rocks Spawning & Throwing: The player can spawn and throw rocks at aliens and continue progressing through the trials.
- Alien Walking: The alien walks around in the trial environment and trying to prevent the player to complete the trials. They stand as obstacles moving rather than actually attack the player.

- Object Falling: Objects such as rocks fall from sky at different frequency and mass and the player must read where the object will land and move out from that zone.
- Dialogue: Captain Emilie will communicate with John through radio and explains some physics concepts such as gravity, mass, and projectile motion.
- Progression: The player must reach the end of the trial to start the next trial. When the player dies in one of the trials he must repeat the same trial again.
- Health: When zombie touch the player health decreases.

1.2.3 Story & Levels

Story

Eight months ago, Captain Emilie, an astronaut that went on a scientific mission to explore a planet that is newly discovered when scientists detected unusual gravity around it. After Captain Emilie arrived the planet, all communications with her were lost.

To find out what happened, Captain John, a retired astronaut and Emilie's friend travels to the planet to find out what happened and search for Emilie, after landing on the planet, the alien boss talks to him and challenges him.

Then John receives a radio transmission from Emilie saying that she survived for eight months and now she is prisoned by the alien boss where the spaceship is located. The alien boss face appears in the planet and tell John that there are three levels of trials if he passes them, he takes Emilie and leave the planet, if not, John, Emilie, and the spaceship are owned by the alien boss and will remain at the planet forever.

John must survive from aliens walking, avoid falling objects, and complete the trials through a radio communication with Emilie while she explains scientific concepts such as gravity, mass, and projectible motion.

Captain John and Captain Emilie must work together to make John:

- Survive falling objects and avoid aliens
- Fight aliens by throwing objects
- Complete both trials
- Reach the area where Emilie and the Spaceship are located
- Rescue Emilie
- Escape the planet with Emilie

Levels

- Level 1 (The Road): The level where John starts the first trial, and its goal to introduce gravity by making the player observe falling objects from the sky and how they fall. The player completes the trial while avoiding only objects.
- Level 2 (The Alien Road): The level where John starts the second trial. The player tries to complete the trial while avoiding falling objects, and avoiding aliens, as this level focus only on gravity and mass.

- Level 3 (The Maze): After finishing the alien road, the player enters the maze where objects still fall, and aliens are still there, but the width of the environment will be narrowed, environment will be harder, and the player gain the ability to spawn and throw rocks at aliens, as this level focus on gravity, mass, and projectile motion. At the end of that level, John reaches the spaceship, rescue Emilie, and escapes the planet.

1.2.4 Characters

- Captain John (Main Character): A retired Astronaut who travels to the alien planet to find out what happened to Emilie
- Dr Emile: An astronaut who joined a scientific mission to the planet and trapped there for eight months.
- Alien Boss: The leader of the aliens on the planet that made the trials.
- Aliens: Controlled by alien boss that try to prevent John to complete the trials.

1.2.5 Technical Specifications

- Platform: Pc
- Game Engine: Unity
- Assets: Different assets will be used from Unity Asset store, SketchFab, and Mixamo.

1.2.6 Sound Design

Music will be integrated into the game so that the player can have the full experience of the game. All music will follow a science-fictions theme that matches the overall style of the game.

1.2.7 Milestones

1. Create the GDD
2. Create the environment for both trials and the planet
3. Implement the characters, movement, and alien system
4. Implement falling objects and throwing objects system
5. Implement the two levels
6. Creating the cutscenes
7. Implement the dialogue between John and Emile
8. Implement the dialogue for the alien boss
9. Implement the sound music, and sound effects
10. Create the UI
11. Testing the game

1.2.8 Prioritized Features

In my opinion there are features that are must-have and features that are nice-to-have, and in the first prototype, only must-have features should be implemented because must-have features are the core features of the game, while nice-to-have features will only add more value and better experience to the game.

There are features that are must-have like the characters, challenges, movement, falling objects, throwing objects, progression, and the dialogue between John and Emile. While there are features that are nice-to-have like adding more alien types, improve the visual details, effects, and adding fighting system.

2. Technical Implementation

2.2 Unity Features

2.2.1 Prefabs

Developers and game creators use prefabs to store game objects in it, as it is a reusable template of game objects, as we can use these prefabs to spawn objects stored in it, and use the same prefab in multiple scenes which reduces time of setting a game object each time we create a scene.

Prefabs were used in Lost Signal for multiple things such as objects that are spawned such as aliens, falling balls, and rocks, their instantiation points in the scene, and main player character.

Prefabs helped me to spawn objects without they're being in the scene before, as it allowed me to use the same alien or main player character prefab to spawn or add them in different scenes without having to add them and configure them in each scene hierarchy.

Prefabs reduced repetitive work of configuring objects in different scenes, as it helped me to reuse the objects in different scenes without having to configure them in each scene hierarchy. The one limitation is managing them when there are a lot of prefabs such as instantiation points for each level and for more than one object, but it could be solved through making folder for each level instantiation point.

2.2.2 Physics engine

Physics engine is used to give life to objects, as it offers colliders and rigid body that maintain some aspects of physics logic like interactions between two objects through triggers and collision, and adds gravity to an object through rigid body.

Colliders were used a lot in Lost Signal, I used triggering colliders on empty game object to detect level end to maintain transitions between scenes and level. I also used usual collider for the health system where when the player interacts or touches an alien or falling ball, the health will decrease. Rigid body were used on falling objects and throwable rocks, as falling objects use gravity to go downward while rocks use force to apply projectile motion.

The physics engine was one of the most important systems in Lost Signal because it addressed the educational part of the game which is gravity and projectile motion through rigid bodies that contains gravity and force. Also, it helped me to implement interactions between objects and without these interactions we wouldn't know if

player health is decreasing. Also, the levels are designed as a maze, so the player mustn't go through the wall which was addressed in the collider and rigid body that give physics characteristics for the player and maze. One limitation was found while creating the game which is even if there are a collider and rigid body, the player might go through the wall if he is going fast enough. An alternative for the gravity and projectile motion would be using translate to move the objects downward as falling or forward, but I would not recommend it since there is a dedicated physics system.

2.2.3 Animation System

Animation system including Animatior makes the game object make a specific animation through Animator controllers, and it lets u control the transitions between animations through states and parameters, as you could implement walking, running, and talking animations through them.

It was used in Lost Signal for two main aspects. The first was the movement, as even though the game is a first person game and the player will not see the movement but I implemented to make the game looks realistic, as in Lost Signal when you move the mouse down you can see the legs and hands of the player moving slow when walking slow, moving natural when walking, and moving faster when running. The second is the cutscenes, as I implemented talking and idle animations and controlled them whenever a character speaks.

The animation system improved the realism of the game, as player movement, and cutscenes appeared realistic, as it allowed me to use parameters and state transitions to actually move between walking and running for the movement, and talking and idle for the cutscenes which resulted in a realistic movement and cutscenes for Lost Signal. One limitation is the complexity, as it can become complex to handle when there are a lot of animations for a specific character. An alternative would be a time-consuming alternative which is handling animations through scripts such as if the player wants to move the hand, the developer must access the object of that hand and move its position, but I would not recommend it since there is a dedicated animation system.

2.2.4 UI System

The UI system is used to display things to the player through UI components such as images, texts, buttons, sliders, which all used for example for displaying health bar, main menus, or subtitles.

UI system was used in Lost Signal to create the main menu screen, game over screen, credits screen, and the components of the UI system primarily buttons and text were used in these screens. I also used UI system in the subtitle part when a dialogue is happening, as I used text for the subtitle and image for the subtitle background.

Without UI system, I would not be able to create an important screen for Lost Signal, as it made me use its components easily to create those screens, health bar that the player without it can't track his health, and the subtitle and they all improved the usability and experience of the game. One limitation I faced is that when the canvas is

an overlay, you cannot change its position, so I had to change the position of the whole world so that its borders don't show.

2.2.5 Audio System

The audio system is used to play music, dialogue, and sound effects through Audio Source and Audio Listener components so that the developer manages some configuration of the audio implemented.

It was used in Lost Signal for background music, as there is music that plays throughout the game in a low volume, and it was used for dialogue whenever there is a cutscene or a dialogue mid-game such as the radio communication between John and Emilie.

Audio system makes the game more engaging, as without it, the game would stay silent which is boring. Audio system in Lost Signal made the game more engaging and enjoyable by adding dialogue for storytelling and learning and adding a background music that makes the player feel the atmosphere of the game. One limitation is that the audio system should lower volume of the audio based on priority, but it doesn't.

2.2.6 Scene Management

Scene Management is a strong feature, as it allows the player to navigate from one level to another and allows the developer to load and unload scene whenever needed which helps us with the game progression.

Scene Management was used to move between the main menu, opening cutscene when it ends, levels when a player reach the end of the level, restart level when player loses, ending cutscene when the player finish the game, and credits scene.

Scene management allowed the game to be divided into multiple separate levels each level operates in specific scene and without it, all scenes would be combined in one scene which is complex to handle, as it made the scenes to be loaded or reloaded when levels are finished or restarted. A limitation is that loading, unloading scenes would introduce some bugs such as freezes or crashes.

2.3 Challenges

During building the game I faced many obstacles and challenges that took time to be solved. The challenge primarily was in implementing some mechanics, assets, and game feel.

Implementing & Debugging Mechanics

Some of the mechanics I implemented through script didn't went as expected such as throw rocks, as the expectation was that the rock go straight forward in a curved line to demonstrate projectile motion, but it appeared that the rock goes forward but not in a realistic way and does not move too far. After searching, I understood that there is something called force mode which demonstrate how force is applied and without explicitly choosing the mode, the force will choose the default mode which applies force gradually through time, but the impulse mode would apply the force instantly

which is the expected from the mechanic. Also, I faced some bugs in the dialogue script where sounds sometimes weren't playing so I printed a word in the console to check what's really happening, but it turned out that I forgot to play the sound in the script.

Assets

One of the challenges I faced was choosing the right assets and the hardest part when finding asset that is not compatible. Some assets I first used took did not match the game theme and take so many triangles and when I say many triangles I mean millions of triangles which would affect the performance of the game, so I tried to choose assets that match the game theme and atmosphere with making sure that they do not take so many triangles. Also, I faced a problem with some assets that the materials are shown as pink then I found out that the material shader does not use universal render pipeline, so I changed it to URP.

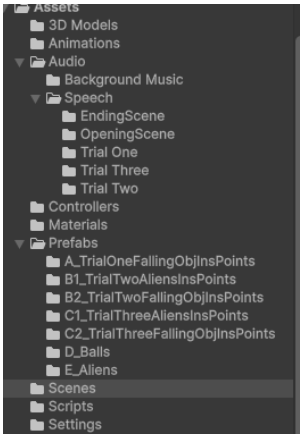
Game Feel

Managing and balancing progression, and levels difficulty was challenging, as some values in spawning objects needed to be adjusted to make the game difficulty feel playable but also not very easy and it was addressed though testing the gameplay each time when adjusting those values. This will also be addressed in the testing part to make sure that the game feel is good.

2.4 Asset Management and Integration

Assets in Lost Signal were used for 3D models, environment, skybox, and background music that were imported from Unity Asset Store. I also used animations, and characters such as John, Emilie, and Alien boss from Maximo.

These assets were organized in structured way by creating separate folders for each category, and each folder might have subfolders for specific scenes and asset types as shown in the picture.



This structure made it easier for me to navigate the project, locate files quickly, reduce confusion and time.

2.5 Implementation of GDD Features

Mechanics

The GDD proposed different mechanics including movement, falling objects, rock spawning and throwing, health, level progression, and dialogue.

- Movement: Implemented successfully through different animations such as walk and run with updating the position of the player each time he presses up arrow or w buttons. The rotation was implemented on the player and the camera, as the player can rotate on Y and the camera rotates on Y and X while X is locked from -90 to 90. Left shift makes the player run, and left control makes the player walk slowly.
- Falling Objects: Implemented successfully through instantiation, as I used it and repeatedly instantiate falling objects at a predefined point.
- Aliens: Aliens was implemented through instantiation, as I used it and repeatedly instantiate Aliens at a predefined point. After instantiation, they use an animation called scary walk to walk forward and when they reach the wall they get destroyed.
- Rock spawning and throwing: Implemented successfully through instantiation, as I used it and repeatedly instantiate rocks in the middle of the camera, but only when the player presses the left click on the mouse. After instantiation they get a force to go straight forward in a curved line.
- Dialogue: Implemented successfully through audio sources and canvas. Synchronizing the audio sources was successful through the Invoke method that allows you to play an audio after a period of time, so I played each audio in the time where they should be played. I also used the canvas for subtitles and used set active to only make it viewable when the audio plays. The audio was generated using platforms like TTS maker and Narakeet.
- Progression & Health bar: Implemented successfully through detectors and scene manager. Detectors were used to detect if the player finished a level and the health bar detect if the player should restart the level. Scene manager was used to load the next scene when detectors detect if the player finished the level.

Story & Levels

The story and levels remain the same and both implemented successfully and exactly as the GDD. The main narrative that involves Captain John, Captain Emilie, and the Alien Bos was successfully implemented and translated into the prototype through cutscenes, and dialogue mid-levels. Dialogue in cutscenes was implemented as shown in the mechanics part but the cutscenes itself used animator for talking and idle to make the scene look realistic.

The levels were also implemented successfully; each level was built in a dedicated scene and the player progress through them through scene manager. The levels educational progress followed a specific path, as level 1 was mainly about only gravity, level 2 was mainly about mass, and level 3 was mainly about projectile motion.

Characters

The characters were implemented successfully. Captain John was implemented as the main playable character, Captain Emilie and the alien boss were implemented through cutscenes and dialogue, and aliens were implemented as obstacles that damage the player when touched.

Technical Specification

The technical specifications remain the same, the game was implemented for pc and developed using Unity. For the assets and 3D models, they were taken from Unity Asset Store and Mixamo. No models were taken from Sketchfab because at first I thought I would take some models from there but appears that all the models needed were found in Unity Asset Store and Mixamo.

Sound Design

The sound design was implemented successfully and closely matches the GDD, as I added background music that were taken from the Unity Asset Store under the name of SCI-FI space tracks.

Overall

All GDD features implemented successfully and accurately in the prototype and all of them match the GDD because I treated the GDD as the foundation of the game before and during development. The gameplay mechanics, story, characters, level progression, technical specifications, and sound design were all translated into the prototype as planned but with a very small change which is using assets only from Unity Asset Store and Mixamo and no using of assets from Sketchfab.

3. Validation

3.1 Playtesting & Feedback Collection

The test group included me, and my brother whose age is 15, a gamer, and like survival and sci-fi games. The test group is small, but the participants represent lost signal target audience, as they are +12, gamers, and like survival or sci-fi games which makes their feedback very useful. The main feedback that I'm focusing to collect is gameplay enjoyment, gameplay engagement, clarity of each level, levels difficulty, and bugs.

Feedback Link: Feedback

3.2 Interpreting Peer Review Feedback

Testing Log

Bug Id	**Description**	**Status**
B1	Stars background in main many were not moving	Fixed
B2	Opening cutscene dialogue mentioned three trials but subtitle mentioned two trials	Fixed
B3	Falling objects did not appear in level 1	Fixed
B4	Subtitle mismatch with Emilie's dialogue in level 3	Fixed
B5	Characters in the ending cutscene played walking animation but did not move	Fixed

Suggestions Log

Feedback Id	**Description**	**Status**
S1	Add crosshair in level 3 to know where we are throwing rocks.	Addressed
S2	Increase difficulty	Addressed
S3	Limit available throwing rocks in level 3	Addressed
S4	Add alien boss fight	Future
S5	Player damage in previous level continues to other levels but with +50 in health	Future
S6	Allow aliens to directly attack the player	Future
S7	Add settings menu	Future

Strengths, weaknesses, and Opportunities

For the strengths, the player mentioned that the game was fun and engaging and they learned gravity clearly. The player also mentioned that the radio communication between Emilie and John has a real value, as the dialogue is one of the most important things that makes a game fun and enjoyable. The player also mentioned levels were clear, as the level design and progression were clear and understandable, as he knew exactly where to go.

For the weaknesses, the player mentioned that levels were easy, as he finished the game without being damaged much, and he mentioned to increase spawn rates or add more aliens and objects which was addressed.

For the opportunities, multiple suggestions were addressed and others were planned to implement in the future. The main suggestions that could be considered as opportunity and is addressed is the crosshair which makes the player knows where he throws the rock. The suggestions that were considered to be implemented in the future are allow aliens to directly attack the player and add settings menu.

4. Refinement & Plan

The prototype demonstrates the core educational objectives, ideas, and core gameplay. The trials are playable and engaging, the falling objects teach gravity and mass, throwing objects teach projectile motion, and the radio communication between John and Emilie provide the educational content through dialogue rather than observation.

The prototype is not a complete game and should be viewed as a foundation instead of a full game. It could be expanded across different areas such as game feel, story, visuals, educational aspect, environment, etc. It also could introduce new different features such as fighting systems, particles, sound effects, etc. to make it a complete game.

Plan

- Game Feel
  This phase focuses on making the prototype feel more complete a through scoring systems that track player performance across the trials including survival times, number of aliens killed, and number of aliens and falling objects avoided, as the score could be displayed at the end of each trial and at the ending screen which give the player a feel of achievement.
  Damage flash effect will be added so that when the player is hit by a falling object or alien, the screen flashes red so the player sees the damage clearly.
  Pause and settings menu will be added to allow players to pause the game anytime they want, adjust audio volume, and mouse sensitivity.
  Another thing will be enhanced is the difficulty of levels to make the game feel more challenging.

- Environment
  This phase focuses on expanding the environment, as the trials zones could be expanded heavily to make it more like an exploration that the player have multiple danger zone and when a zone trial is completed, he makes that a safe zone. Also, I can also add more alien types.

- Story
  This phase focuses on the narrative of the game, as current story is delivered in opening and ending scenes with a little narrative during the trials. So, after expanding the environment, I could add more radio communication through the game. I can add more dialogue for John and Emilie relationship to hit the emotional part of the player. I can also add more alien boss dialogue during or mid trials. Also, I can adjust and enhance story.

- Sound Effects and Particles
  Sound effects will be added for major events, as falling objects will produce impact sound based on their size when they hit the ground, thrown rock will produce a sound when throwing, player movement will produce walking or running effects along with breathing and a sound when the player health decreases. Also, other sound effects will be added such as UI interactions, alien movement, wind, and scene transitions which all make the game feel more alive.
  Particles will be added to major moments, as dust particles will be added when objects hit the ground, player's movement, and aliens' movement.

- Visuals
  This phase focuses on improving the visuals, as I can add more details for the environment by using different assets, improved lighting, and improved textures which allow the game to perform better visually.

- Educational Aspect
  This phase focuses on making the educational aspect bigger and deeper, as the current game focuses on gravity, mass, and projectile motion but I can make it focuses on more physics areas such as energy, electricity, or even waves and sounds which would make the game educational content heavier and could make it a game that teaches most of physics concepts which make it hit more target audiences.
  Also, currently educational dialogue is triggered on a specific place, but I can make it more dynamic through making the educational dialogue plays while the player is experiencing a specific physics aspect.

- Levels
  The last phase focuses on expanding levels and adding a complete fighting system, as I can add more levels to the game or make the current levels environment bigger so that the player explores a part of the planet on each phase.
  Also, I can add a fighting system that make the player combat with aliens after making the environment bigger and more like exploration, as aliens can walk towards the player and player must start a combat with them to keep moving.

References

1. Huston, C. (2023) Game engines, Medium. Available at: https://medium.com/@chusto2/game-engines-8763c6782eef
2. Gregory, J. (2009) Game engine architecture. Available at: https://studio.eecs.umich.edu/confluence/download/attachments/17827062/Game_Engine_Architecture_-_Jason_Gregory.pdf?api=v2
3. Unity Discussions (2022) HDRP vs URP what is the tradeoff? Available at: https://discussions.unity.com/t/hdrp-vs-urp-what-is-the-tradeoff/874512

4. Meshy (2024) What is Unreal Engine? Everything you need to know to get started. Available at: https://www.meshy.ai/blog/what-is-unreal-engine
5. Game Design Skills (2023) Is CRYENGINE a good game engine? Available at: https://gamedesignskills.com/game-development/is-cryengine-good/
6. Meckeys (2023) Game engines and performance: how technology shapes your gaming experience. Available at: https://meckeys.com/hardware/game-engines-and-performance-how-technology-shapes-your-gaming-experience/
7. Steam (2024) Games made with Unity. Available at: https://store.steampowered.com/curator/39750107-Games-Made-With-Unity/
8. Steam (2024) Games made with Unreal Engine. Available at: https://store.steampowered.com/curator/39869360-Games-made-with-Unreal-Engine/
9. Game Design Skills (2023) 48 popular games made with CRYENGINE. Available at: https://gamedesignskills.com/game-design/popular-cryengine-games/
10. Gamixlabs (2023) Evolution of game engines: past to future. Available at: https://gamixlabs.com/blog/the-evolution-of-game-engines/
11. Meegle (2024) Game engine for cloud gaming. Available at: https://www.meegle.com/en_us/topics/game-engine/game-engine-for-cloud-gaming
12. Meegle (2024) Game engine for procedural generation. Available at: https://www.meegle.com/en_us/topics/game-engine/game-engine-for-procedural-generation
13. Meegle (2024) Game engine for virtual reality. Available at: https://www.meegle.com/en_us/topics/game-engine/game-engine-for-virtual-reality
14. T-Plan (2023) AI and machine learning in game development. Available at: https://www.t-plan.com/blog/ai-and-machine-learning-in-game-development/
15. Unity Learn (2020) Lab 1 - Project design document. Available at: https://learn.unity.com/course/create-with-code/unit/unit-1-driving-simulation/tutorial/lab-1-personal-project-plan
16. Unity Technologies (2020) Game design document template. Available at: https://connect-prd-cdn.unity.com/20201215/83f3733d-3146-42de-8a69-f461d6662eb1/Game-Design-Document-Template.pdf
17. Altgilbers, I., Berlin, J., Cooper, A., Gustavson, E., Harris, G., Knowles, M., Ngu, H., Scott, G., Singhal, R., Stewart, E., Thayer, D., Verola, L., Zhao, S. and Francois, A. (2009) An Ant's Life: game design document. Available at: https://gamescrye.com/wp-content/uploads/2016/08/AnAntsLife-GameDesignDocument.pdf

Tutorials, Forums, Documentations

1. Infallible Code (2020) First person movement in 10 minutes - Unity tutorial, YouTube. Available at: https://www.youtube.com/watch?v=f473C43s8nE
2. Brackeys (2019) How to make a simple health bar in Unity, YouTube. Available at: https://www.youtube.com/watch?v=0tDPxNB2JNs
3. Unity Learn (2019) Lesson 5.3 - Game over. Available at: https://learn.unity.com/tutorial/lesson-5-3-game-over-1

4. Natty GameDev (2023) Uncover the simple trick to keep music playing between scenes, YouTube. Available at: https://www.youtube.com/watch?v=xswEpNpucZQ
5. Unity Discussions (2018) How do I create an exit/quit button? Available at: https://discussions.unity.com/t/how-do-i-create-a-exit-quit-button/142125
6. Unity Technologies (2024) Unity scripting API: SceneManager. Available at: https://docs.unity3d.com/6000.4/Documentation/ScriptReference/SceneManagement.SceneManager.html
7. Unity Discussions (2019) Unlock cursor. Available at: https://discussions.unity.com/t/unlock-cursor/167385
8. Unity Discussions (2021) How to freeze and unfreeze my game? Available at: https://discussions.unity.com/t/how-to-freeze-and-unfreeze-my-game/311091
9. Unity Discussions (2022) InvokeRepeating a random audio clip every x seconds. Available at: https://discussions.unity.com/t/invokerepeating-a-random-audio-clip-every-x-seconds/811412
10. Comp-3 Interactive (2021) How to pick up, drop, and throw objects in Unity, YouTube. Available at: https://www.youtube.com/watch?v=FnE4aS0dsE4
11. Unity Discussions (2020) Throwing objects. Available at: https://discussions.unity.com/t/throwing-objects-694476/694476

Speech Generation Platforms

1. TTSMaker (2024) Free text to speech online. Available at: https://ttsmaker.com/
2. Narakeet (2024) Easily create voiceovers and narrated videos using realistic text to speech. Available at: https://www.narakeet.com/app/text-to-audio/

Assets

1. Unity Asset Store (2019) Lunar Landscape 3D. Available at: https://assetstore.unity.com/packages/3d/environments/landscapes/lunar-landscape-3d-132614
2. Unity Asset Store (2024) Maze Modular Puzzle Kit. Available at: https://assetstore.unity.com/packages/3d/environments/maze-modular-puzzle-kit-302221
3. Unity Asset Store (2021) FREE Skyboxes - Space. Available at: https://assetstore.unity.com/packages/2d/textures-materials/sky/free-skyboxes-space-178953
4. Unity Asset Store (2022) Alien (Different Skins) and Glowing Sword. Available at: https://assetstore.unity.com/packages/3d/characters/creatures/alien-different-skins-and-glowing-sword-215878
5. Unity Asset Store (2024) Arachnid-Alien Head Lite. Available at: https://assetstore.unity.com/packages/3d/characters/creatures/arachnid-alien-head-lite-335052
6. Unity Asset Store (2022) UI SFX Free Pack. Available at: https://assetstore.unity.com/packages/audio/sound-fx/ui-sfx-free-pack-245925

7. Unity Asset Store (2024) Free 30 Sci-Fi Space Tracks Music Pack. Available at: https://assetstore.unity.com/packages/audio/music/free-30-sci-fi-space-tracks-music-pack-354566
8. Audiomos (2022) Sci-Fi buttons SFX pack, YouTube. Available at: https://www.youtube.com/watch?v=XD3K7X7OlR0
9. Unity Asset Store (2018) HQ Rock Pack Free. Available at: https://assetstore.unity.com/packages/3d/props/exterior/hq-rock-pack-free-83388
10. Flaticon (2024) Crosshair icons and symbols. Available at: https://www.flaticon.com/free-icons/crosshair